THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 2: **Exploring Mixtures and Their Separation** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A pure substance consists of: 1

(a) a single kind of particle with fixed composition
(b) only metals
(c) only gases
(d) two or more substances in any ratio

2. A suspension is: 1

(a) a pure substance
(b) a heterogeneous mixture whose particles settle on standing
(c) a clear solution
(d) a compound

3. In paper chromatography the pencil line is drawn at the bottom because: 1

(a) pencil is faster **(b)** pencil does not dissolve in the solvent
(c) ink is better **(d)** it looks neat

4. Which of the following is a pure substance? 1

(a) sea water **(b)** air **(c)** distilled water **(d)** milk

5. Which of the following is a colloid? 1

(a) sugar in water **(b)** salt in water **(c)** fog **(d)** air

6. In chromatography the component which is more soluble in the solvent: 1

(a) disappears **(b)** stays at the base
(c) moves down **(d)** moves farther up the paper

7. Which of the following is a homogeneous mixture? 1

(a) oil and water **(b)** muddy water **(c)** brass **(d)** sand and sugar

8. Particles of a colloid cannot be separated by: 1

(a) centrifugation **(b)** ordinary filtration
(c) dialysis **(d)** coagulation

9. Alloys are: 1

(a) heterogeneous **(b)** pure metals
(c) compounds **(d)** homogeneous mixtures of metals

10. A mixture of sand and iron filings is: 1

(a) homogeneous **(b)** a pure substance **(c)** heterogeneous **(d)** a compound

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Differentiate between a homogeneous and a heterogeneous mixture with one example each. 2

12. What is centrifugation? Give one use. 2

13. Calculate the mass by mass percentage of a solution having 25 g of sugar in 75 g of water. 2

14. Define solubility. How does it vary with temperature for a solid in water? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the method of separating a mixture of salt, sand and ammonium chloride. 3

16. How much solute is needed to prepare 250 g of a 12 per cent (m/m) sugar solution? 3

17. What is meant by a saturated solution and an unsaturated solution? How can we convert one to the other? 3

OR

Describe how crystallisation is used to purify an impure sample of copper sulphate. 3

18. How would you prepare 200 mL of a 15 per cent (v/v) solution of alcohol in water? 3

OR

Explain why we must not evaporate to dryness while crystallising a solution. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Differentiate between pure substances and mixtures. Classify each as pure, homogeneous mixture or heterogeneous mixture: air, sugar solution, gold, milk, brass. 5

OR

Describe the working and use of paper chromatography with the diagram described in words, including R_f ideas. 5

20. Describe the properties of true solutions, suspensions and colloids using three activities. 5

OR

Explain the process of crystallisation and its advantages over evaporation. State two applications. 5

END OF QUESTION PAPER Total: 40 Marks

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